In the popular imagination, Einstein lost the Bohr-Einstein debates. We are told that Einstein, a creative revolutionary in his youth, had grown old and conservative, and was unable to accept or even understand the dramatic implications of the new quantum theory. (At the time of the Solvay Conference Einstein was forty-eight years old.) Physics subsequently went on without him, as the great man retreated to pursue idiosyncratic attempts at finding a unified field theory.

Nothing could be further from the truth. While Einstein failed to put forward a complete and compelling generalization of quantum mechanics, his insistence that physics needs to do better than shut up and calculate was directly on point. It is wildly off base to think that he failed to understand quantum theory. Einstein understood it as well as anyone, and continued to make fundamental contributions to the subject, including demonstrating the importance of quantum entanglement, which plays a central role in our current best picture of how the universe really works. What he failed to do was to convince his fellow physicists of the inadequacy of the Copenhagen approach, and the importance of trying harder to understand the foundations of quantum theory.

If we want to follow Einstein's ambition of a complete, unambiguous, realistic theory of the natural world, but we are discouraged by the difficulties of tacking new hidden variables onto quantum mechanics, is there any remaining strategy left?

One approach is to forget about new variables, throw away all the problematic ideas about the measurement process, strip quantum mechanics down to its absolute essentials, and ask what happens. What's the leanest, meanest version of quantum theory we can invent, and still hope to explain the experimental results?

Every version of quantum mechanics (and there are plenty) employs a wave function or something equivalent, and posits that the wave function obeys Schrödinger's equation, at least most of the time. These are going to have to be ingredients in just about any theory we can take seriously. Let's see if we can be stubbornly minimalist, and get away with adding little or nothing else to the formalism.